Multi-timescale adaptive behavior in a dynamic patch foraging environment

Summary: Natural decisions unfold through continuous interaction between an organism and its environment. Foraging is an innate yet flexible behavior where animals repeatedly decide whether to exploit a depleting patch or leave to explore alternatives. We developed a head-fixed virtual reality patch foraging task that preserves the hierarchical and self-paced structure of natural behavior while allowing experimental control and access. Mice traversed a virtual linear track and made self-initiated choices to harvest probabilistic water rewards at odor-cued sites. Odors cued three patch types: high- and low-value patches whose reward probabilities depleted with each collected reward, and an unrewarded patch. Naive mice rapidly acquired the task, learning to use odor identity and reward outcomes to guide patch-leaving decisions in fewer than seven sessions on average. We used multi-timescale hidden markov models to identify hierarchically structured motifs in running velocity and respiration. Within a patch, run, approach, stop, and leave motifs occurred in sequence, and shifted as mice approached patch leaving, reflecting a latent decision variable. Individual mice had characteristic engagement state dynamics across sessions. Mice left both rewarded patch types at equivalent reward probabilities, consistent with the Marginal Value Theorem's (MVT, Charnov 1976) prediction to leave when local reward rate falls below the global average, but inconsistent with simpler heuristics such as reward counting or win-stay/lose-shift strategies. Independent manipulations of travel cost or global reward rate shifted patch-leaving decisions as predicted by MVT, demonstrating that mice use global reward rate to set local patch leaving thresholds. A Bayesian MVT (McNamara and Houston 1980) with two timescales of learning reproduced mouse behavior in steady-state and dynamic conditions. This task offers an avenue to investigate brain-wide circuit dynamics for adaptive behavior anchored in normative foraging theory.

Additional Detail:

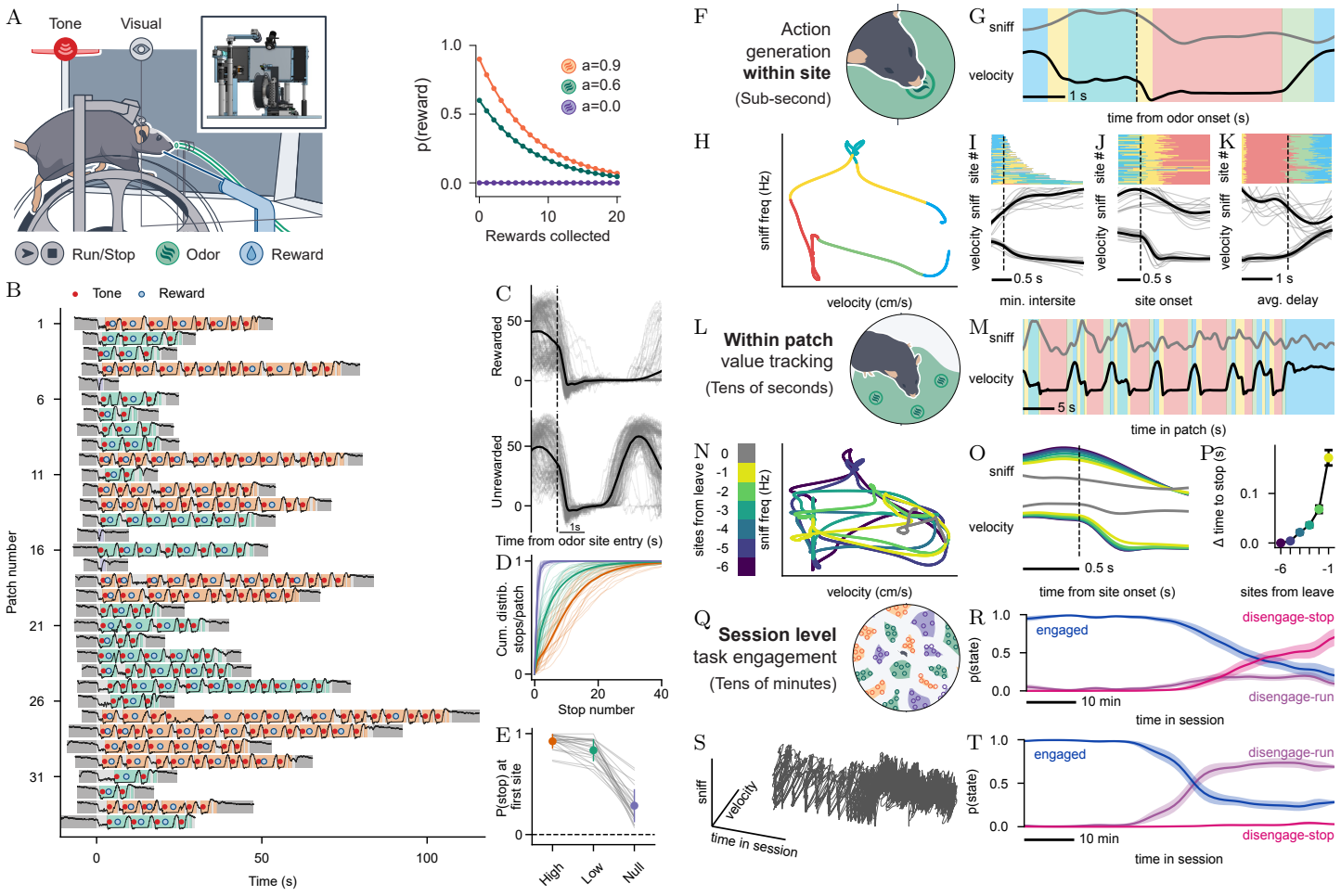


Figure 1: Multi-timescale structure of behavior in a virtual patch-foraging task. (A) Left: head-fixed mice run on a wheel through a multi-sensory virtual odor patch foraging environment. Right: High, low, and unrewarded patch types are cued by odors, differ in initial reward probability, and depletes with each reward collected. (B) Running velocity (black trace), stop choice tones, rewards, and odors (orange, green, or purple) for example patches in a session. (C) Running velocity shows stereotyped stop actions at odor site entries and longer stop times in rewarded vs unrewarded sites. (D,E) Mice foraged in two rewarded patch types, stopping more in the high value type, and consistently skipping the unrewarded patch type (across mice, n=24) (F-T) Behavior is organized across nested timescales: sites (sub-second), patches (tens of seconds to minutes), and sessions (tens of minutes). (G-K) Stereotyped action motifs at the odor site level. Velocity and sniffing in an odor site shaded by HMM states (G) and as trajectory (H). (I–K) HMM states (one row per site) and mean velocity aligned to odor site approach state (I), site entry (J)and leaving (K). (M-P) Action motifs shift as mice approach patch leaving decision. HMM states (M) and velocity-sniff trajectories (N) colored by sites from patch leaving. (O,P) Mice enter odor sites with decreased sniff frequency, increased velocity, (O), and slower reaction times (P; mean ± s.e.m., $n = 13$ mice). (R,T) Session-averaged probability of three engagement states (engaged, disengage-run, disengage-stop; mean ± s.e.m. across sessions) for two mice, one disengaged by stopping (R, $n = 20$ sessions) the other disengaged by running (T, $n = 18$ sessions). (S) Velocity and sniff frequency reflect engagement across time in an example session.

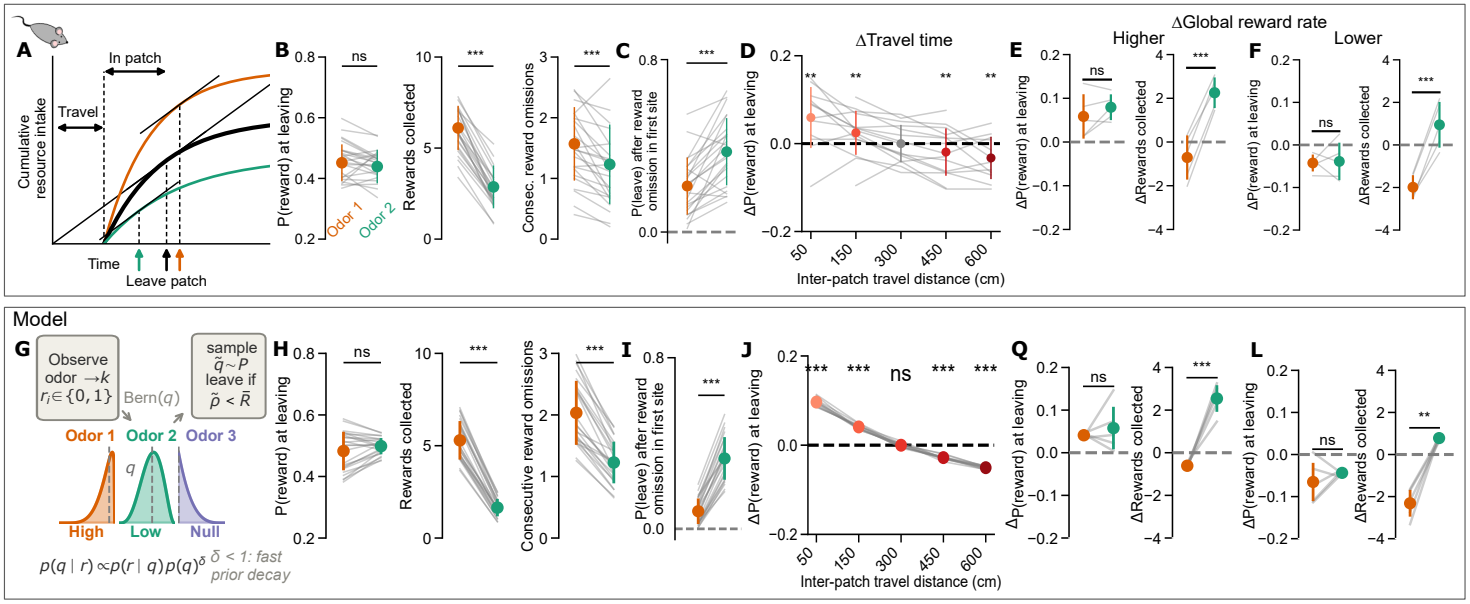


Figure 2: Patch-leaving behavior is consistent with a Bayesian formulation of the Marginal Value Theorem. (A) MVT predicts that optimal patch leaving is set by the global reward rate (slope of black tangent lines), and foragers should leave high or low value odor patch types (orange and green lines) at the same local reward rate. (B) Reward probability at leaving is matched across patch types, while rewards collected, and number of reward omissions experienced differed significantly. Lines are individual mice ($n = 24$). (C) Mice have odor-cued expectations about patch quatlity. Given that the first odor site in a patch was unrewarded, mice were more likely to leave the low value patch (green) than the high. (D) Change in P(reward) at leaving relative to control as a function of inter-patch travel distance. Longer distances led mice to leave at lower reward probabilities, while shorter distances produced the opposite effect, consistent with MVT predictions. (E,F Experiments manipulating global reward rate. In (E), odor 2 reward statistics were increased to match those of odor 1; which led to an increase in global reward. Relative to control sessions, both patch types shifted to higher reward probabilities (left), meaning mice left patches earlier. To achieve this, rewards collected shifted in opposite directions across patch types (right) .In (F) Opposite manipulating where we decreased the global reward rate. Both patch types shifted to lower reward probabilities meaning the mice left patches later. Both results are qualitatively compatible with MVT. (G) Bayesian MVT model schematic. The agent maintains a separate posterior over reward probability q for each odor-cued patch type, updated after each binary outcome via Bayes' rule with a forgetting parameter $\delta < 1$ that discounts accumulated prior beliefs at each update step. At each odor site, a sample $\tilde{q}$ is drawn from the current posterior; the agent leaves if the implied local reward rate falls below a running estimate of the global reward rate $\bar{R}$. (H–L) Bayesian MVT model simulations reproducing the mouse results in (B–F): reward probability at leaving, rewards collected and consecutive reward omissions (H), probability of leaving after early consecutive omissions (I, change in P(reward) at leaving with inter-patch distance (J), and results for higher (Q) and lower (L) global reward rate environments. Significance (paired t-tests across subjects): ns, not significant; $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.